

Session 1 - Defining and Collecting Data

!!!!!!!

Defining Variables

Classification of Variables

1. Categorical Variables (Qualitative) - All about category, ciri khasnya adalah sesuatu yang diukur akan berbentuk kata, walaupun hasil ukurnya mungkin berupa angka. Biasanya bersifat subjektif.

contoh: Partai PDIP(30%), Gerindra(10%) dan seterusnya.. Grade Beef A(202 orang), B(10 orang), C (3 orang) .

2. Numerical Variables (Quantitative) - All about number, ciri khasnya ada angkanya di sesuatu yang diukur. Urutan di dalam grafik atau tabel harusurut agar mudah dipahami

contoh: Maba BINUS umur 16(5%), Umur 17(45%), Umur 18(30%), dst.

- Discrete Variables (*usually counted one by one*)
- Continuous Variables (*usually counted using measuring instruments*)

Measurement Scale & Type

Type	Scale	Characteristic	Example	Math Symbol
Categorical	Nominal	Identifikasi jumlah dan klasifikasi obyek	Jenis kelamin	$\neq$
Categorical	Ordinal	Jumlahnya mengindikasikan posisi relatif obyek lain tetapi bukan berarti perbedaan diantara mereka	Urutan pemenang pertandingan lari	$\neq, > <$
Numerical	Interval	Perbedaan antara obyek-obyek dapat dibandingkan, titik nol relatif	Temperatur	$\neq, > <, + (-)$
Numerical	Ratio	Ada titik nol mutlak, panjang nilai-nilai skala ratio, bobot dpt dihitung	Panjang, bobot	$\neq, > <, + (-), \times :$

No. Restaurant	Quality Rating	Meal Price
1	Good	1
2	Very Good	2
3	Good	2
4	Excellent	3
5	Very Good	3
6	Good	2
7	Very Good	1
8	Very Good	1
9	Very Good	2
10	Good	1

Variable *Quality rating* adalah variable kategori

Variable *Meal Price* adalah variable numerik

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Collecting Data

Data collected from sample, as a part of a population.

POPULATION

A population contains all the items/individuals of interest that the data is wanted to be collected.

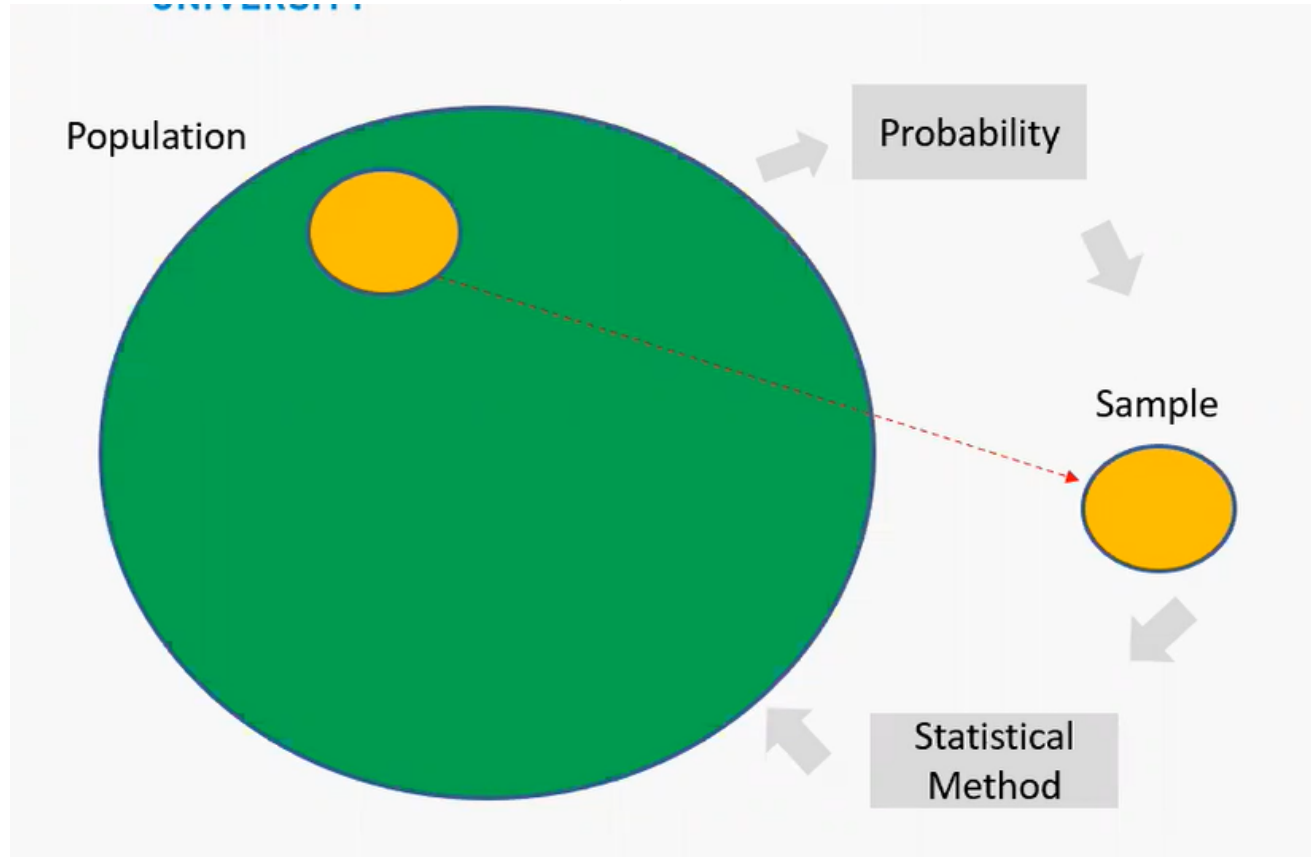
SAMPLE

Sample contains only a portion of population (**not outside of it**).

A sample is a representation of a population.

Sample characteristic needs to be the same with their corresponding population for the data to be valid.

Ketika kita menganalisa data sampel kita menggunakan analisa statistik



Misal: dari 2000 orang diambil 100, maka analisa statistik dari 100 orang...

Analisa statistiknya adalah apakah sampel berasal dari populasi atau tidak

Untuk menilai apakah suatu sampel representatif dari sebuah populasi, digunakanlah inferensial (probability).

Types of Sampling

!!!!!!!

"Types of Survey Error" is not created yet. Click to create.

Session 2 & 3

Organizing and Visualizing Variables - YouTube

Summary table: 2D, misalnya kandungan karbo dalam produk

Kontingensi: 3D, kayak misal

!!!!!!!

Data Visualization

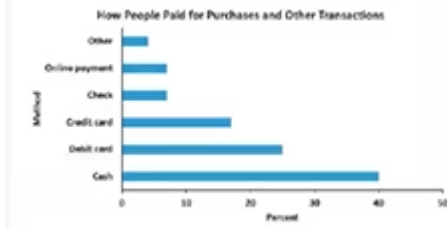
Categorical Visualization

ONE CATEGORY VARIABLE

- Bar Chart 📊

Easier to understand

Excels in comparison (financial statement, etc.), dollars and others.

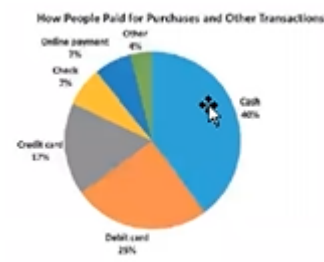


- Pie Chart

Excels in percentage data visualization

Easy to understand for more general people

If there is too many category, or too many little percentages, it might not translate well because it become indistinguishable and unlegible (the category and percentage becomes too small).



- Doughnut Chart

Pie chart with hole in the middle

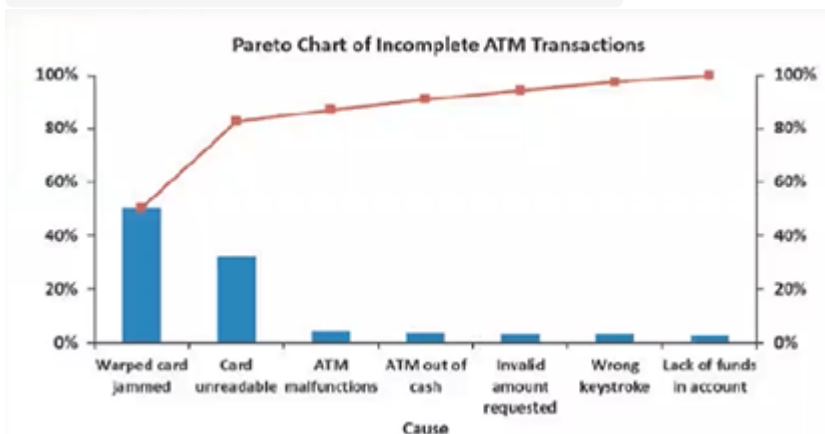
Arguably harder to read

- Pareto Chart

Harder to understand for general people because it introduce new variable graph (the cummulative line) into the mix. Also maybe hard to send your messages across.

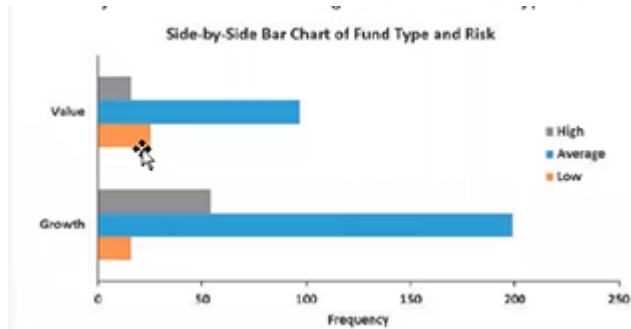
Very complete because you can also the data relative to other category using the cummulative line.

in Ms. Excel use combo icon in the insert tab



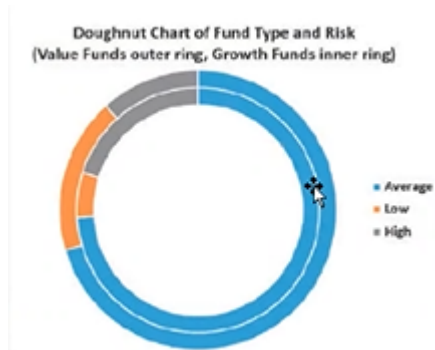
- Side by Side

The modified bar chart to fit more than one variable



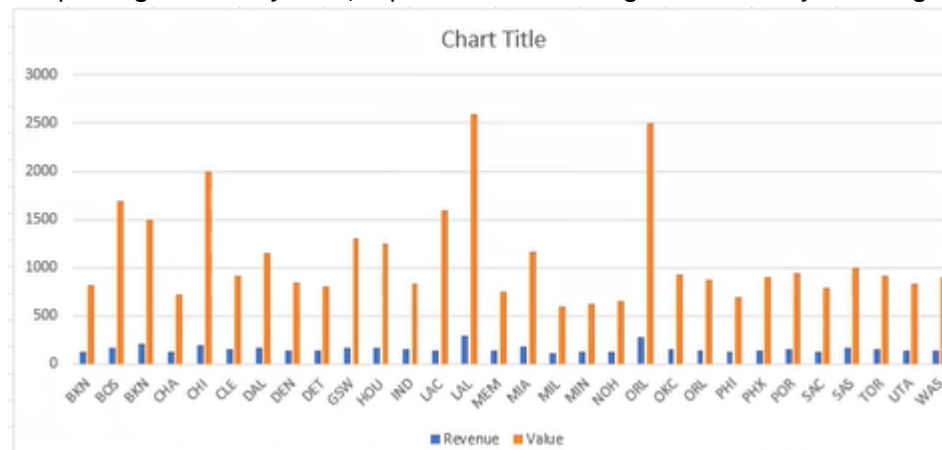
- Doughnut Chart

Pie chart can't be manifested with 2 var.. doughnut is a nice alternative



- Bar chart

Mirip dengan side by side, tapi dibalik dan dengan lebih banyak kategori



Numerical Visualization

ONE NUMERIC VARIABLE

- Stem n leaf

from the data itself, it creates some sort of bar chart. It visualizes data by presenting data as one/more row-wise stems that represent a range of values where each have *leaf* that

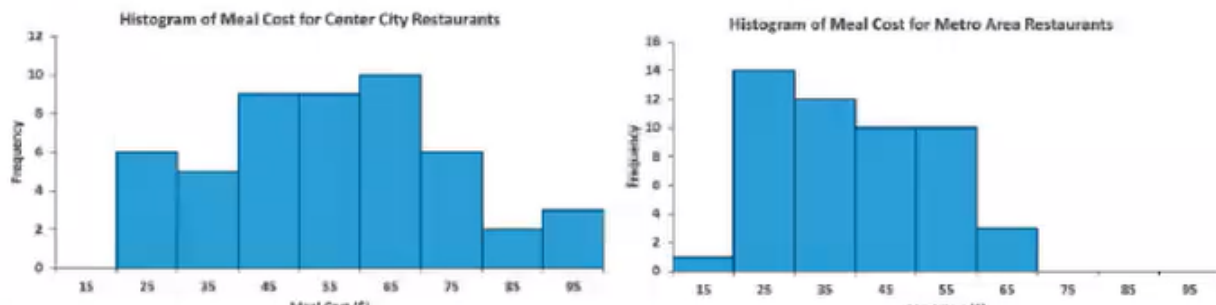
connects to it.

```
2 | 0
3 | 025
4 | 11378
5 | 133467889
6 | 024559
7 | 147
8 | 8
9 |
10 | 2
```

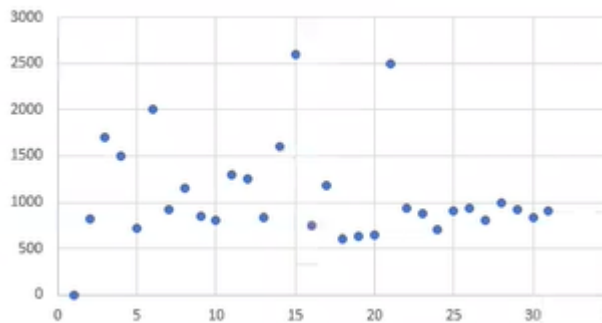
- Histogram
Bar chart, but the variable is a range of value, per kelasnya dan frekuensinya.

FIGURE 2.8

Excel frequency histograms for meal costs at center city and metro area restaurants



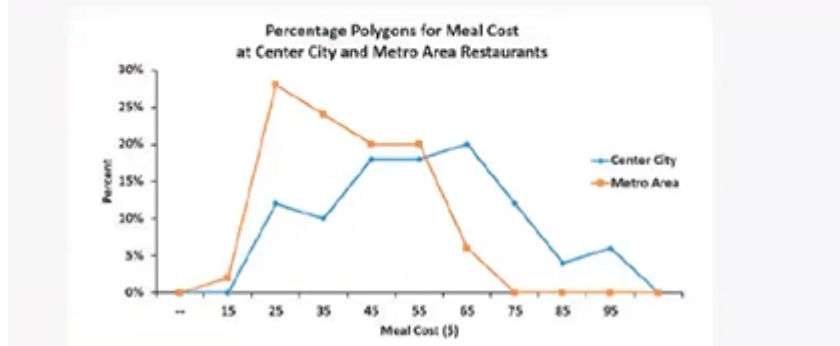
- Scatter



- Percentage Polygon
Jika variable numerik dibagi menjadi 2 menurut kategorinya, bisa digunakan percentage

polygon, dan ada jg byk jenis polygon lainnya

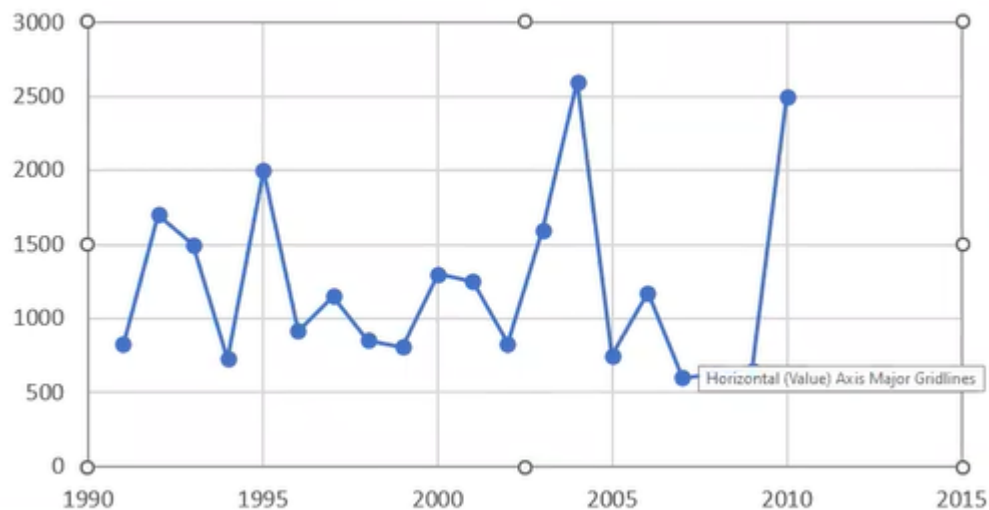
When using a categorical variable to divide the data of a numerical variable into two or more groups, you visualize data by constructing a **percentage polygon**. This chart uses the midpoints of each class interval to represent the data of each class and then plots the midpoints, at their respective class percentages, as points on a line along the X axis.



TWO CATEGORY VARIABLE

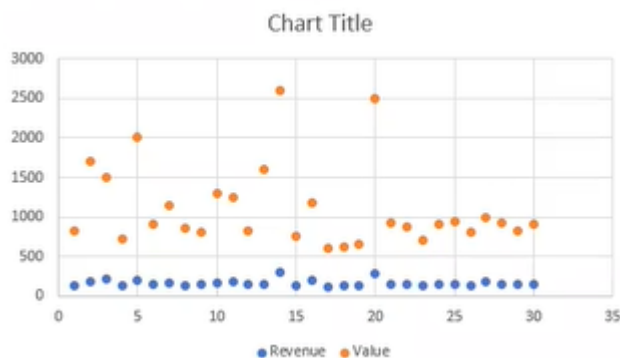
- Time Series

To-view development and trends for specified amount of time



- Scatterfloat

To see patterns and correlation between 2 numerical variables



!!!!!!!

Organizing data for Categorical and Numerical Variables are different.

TABLE CONTINGENCY

`for minimal 2 or more variable`

Table contingency is a good organizing method for categorical variable. Don't forget to add the total row and column.

Table Contingency

Quality Rating	Meal Price			Total
	10-19 \$	20-29 \$	30-39\$	
Good	2	2		4
Very Good	2	2	1	5
Exce lent	0	0	1	1
Total	4	4	2	10

SUMMARY TABLE

`for 1 variable only`

The category is in the left, the percentage or frequency is on the right.

A **summary table** tallies the set of individual values as frequencies or percentages for each category. A summary table helps you see the differences among the categories by displaying the frequency, amount, or percentage of items in a set of categories in a separate column.

Method	Percentage
Cash	40%
Debit card	25%
Credit card	17%
Check	7%
Online payment	7%
Other	4%

Source: Data extracted and adapted from "Paper beats Plastic," *USA Today*, December 26, 2014, p. 1A.

Table 2.1
How People Pay for
Purchases and Other
Transactions

Example

Hago Knife Hit

Sebelum di organisasikan, low-form data

Game	Lawin	Hasil	Store
1	Baru	win	3
2	Baru	win	3
3	Lama	win	3
4	Baru	win	3
5	Baru	win	3

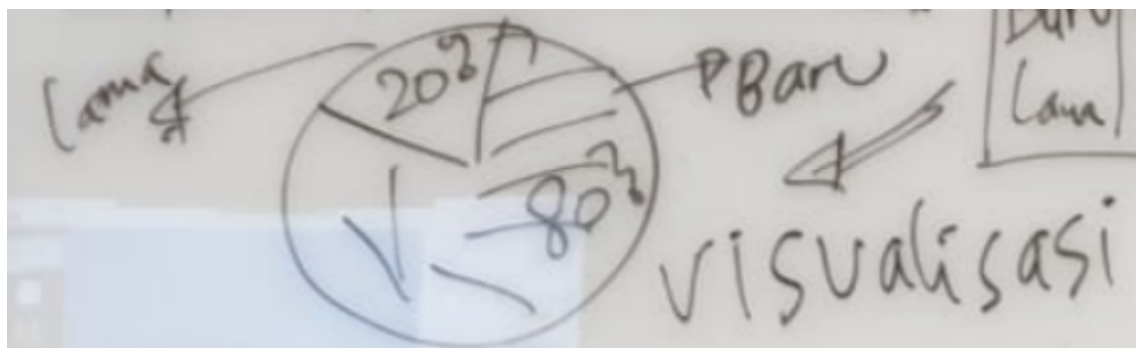
Setelah diorganisasikan

Hasil

		Win	Lose	draw	Total Game	Total Score
LAWAN	BARU	4	0	0	4	12
	Lama	1	0	0	1	3
	Total Game	5	0	0	5	
	Total Score.	15	0	0		15.

Data yang di organisasikan lalu dapat di visualisasikan. [Data Visualization](#)

	F	Paket
Baru	4	9/500 = 80%
Lama	1	5/500 = 20%



Karena orang-orang butuh data yang sudah di proses.

!!!!!!!

Organizing Numerical Data

Preparing before making a Frequency Table

What needed to be decided first?

- Maximum data (highest value)
- Minimum data (lowest value)
- Number of class

Jika data banyak ditentukan dengan [Sturges' Rule - statology.org](http://statology.org)

Jika data sedikit, di kira-kira saja.

- Interval Kelas

DETERMINING THE CLASS INTERVAL WIDTH

$$\text{Interval width} = \frac{\text{highest value} - \text{lowest value}}{\text{number of classes}}$$

Making a Frequency Table

1. Search maximum and minimum data by using the sort function in excel
2. Find the class interval using the decided number of class, highest value and lowest value.
3. Make the range of classes one by one from the minimum + interval, etc.
4. Search the *frequency* for each class and the total frequency. (by the tally)
5. Search the percentage, by *frequency/total*. The total should be 100
6. Search the *relative frequency* and *relative percentage*, by *frequency/total*
7. Search the *frequency cummulative* and *percentage cummulative*

TABLE 2.12

Developing the Cumulative Percentage Distribution for Center City Restaurant Meal Costs

From Table 2.10:		Percentage of Meal Costs That Are Less Than the Class Interval Lower Boundary	
Class Interval	Percentage		
20 but less than 30	12%	0	(there are no meals that cost less than 20)
30 but less than 40	10%	12%	= 0 + 12
40 but less than 50	18%	22%	= 12 + 10
50 but less than 60	18%	40%	= 12 + 10 + 18
60 but less than 70	20%	58%	= 12 + 10 + 18 + 18
70 but less than 80	12%	78%	= 12 + 10 + 18 + 18 + 20
80 but less than 90	4%	90%	= 12 + 10 + 18 + 18 + 20 + 12
90 but less than 100	6%	94%	= 12 + 10 + 18 + 18 + 20 + 12 + 4
100 but less than 110	0%	100%	= 12 + 10 + 18 + 18 + 20 + 12 + 4 + 6

		tabel distribusi frequency							
	Revenue								
Team Code	Revenue		kelas	tally	freq	percentage	Freq Relatif	Percen relative	Freq cummulative
1 BKN	212	129.5	130 - 146.4	1111	4	40	0.4	0.4	4
2 CHI	201		146.5 - 163.4	1	1	10	0.1	0.1	5
3 BOS	173		164- 180.4	111	3	30	0.3	0.3	8
4 DAL	168		181 - 197.4		0	0	0	0	8
5 GSW	168		198 - 214.4	11	2	20	0.2	0.2	10
6 CLE	149				10	100			
7 DET	144								
8 DEN	136								
9 BKN	133								
10 CHA	130								
max	212								
min	130								
kelas 5									
interval kelas	16.4								

!!!!!!!

Probability

Peluang $P(A) = \text{Kejadian} / \text{Total Kejadian}$

Probability of Occurrence

$$\text{Probability of Occurrence} = \frac{X}{T}$$

Where

X = Number of ways in which the event occurs

T = Total number of possible outcomes

Event contoh: muncul angka 1

Suatu kejadian yang diukur peluangnya untuk terjadi

Sample Space contoh: yang himpunan

Ruang sampel, banyaknya kejadian munculnya warna

Event Joint

Suatu kejadian yang meningkatkan/menurunkan peluang dari suatu kejadian, jadi ada 2 kejadian yang diukur

Himpunan kosong = Impossible event, tidak mungkin terjadi

Total Kejadian $T = n(A)$

$A = \text{muncul angka 1}$

$$A = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (3,1), (4,1), (5,1), (6,1)\}$$

$$n(A) = 11$$

$$P(A) = \frac{11}{36}$$

Komplemen A'

Semua yang di ruang sample yang bukan merupakan kejadian

$$P(A) = 1 - P(A')$$

$$A' = 1 - P(A)$$

$$= 1 - \frac{11}{36} = \frac{25}{36}$$



$B \subseteq A = B$ anggota dari A

$A = \text{muncul angka 1}$

$$A = \{(1,1), (1,2), (1,3), (1,4), (1,5)^*, (1,6), (2,1), (3,1), (4,1), (5,1)^*, (6,1)\}$$

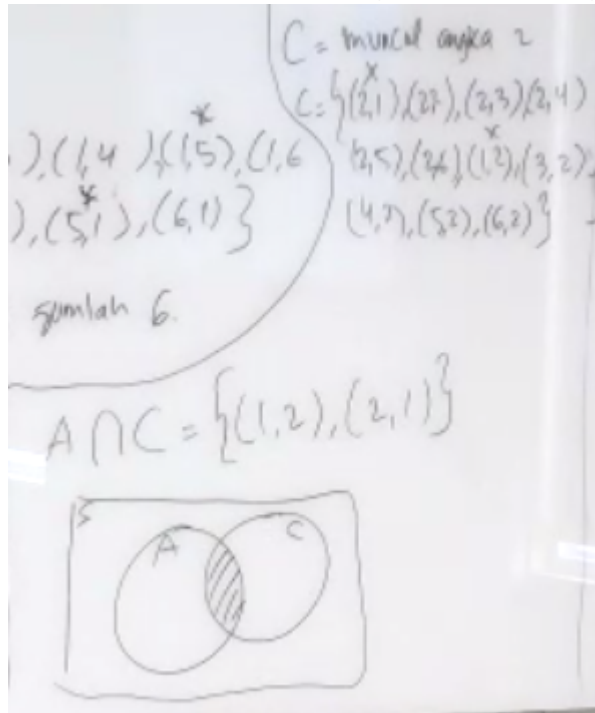
$B = \text{muncul angka 1 dan jumlah 6}$

$$B = \{(1,5), (5,1)\}$$

$B \subseteq A$

$A \cap C$ = Irisan antara A dan C (and)

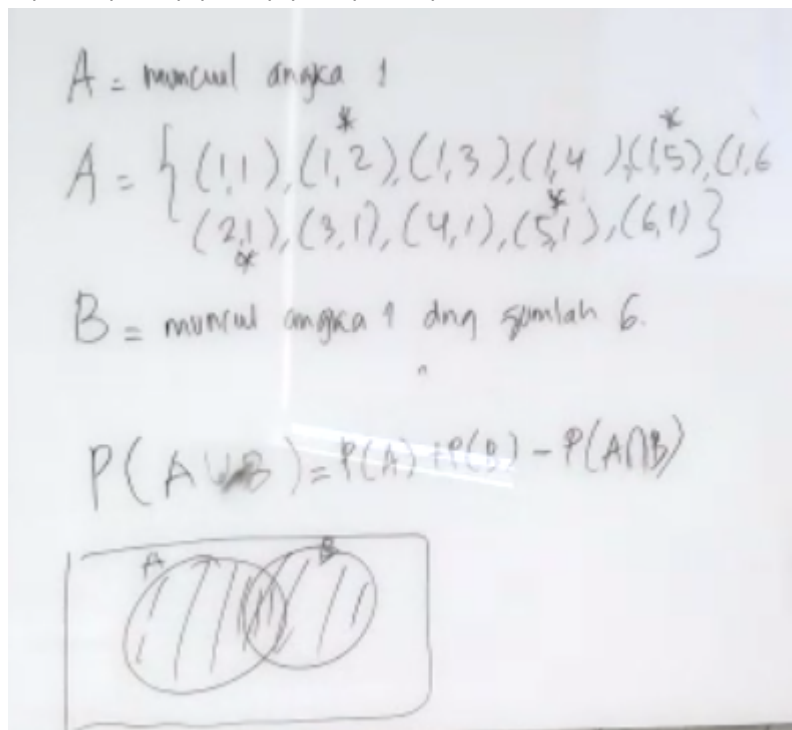
hasilnya adalah elemen yang merupakan anggota A dan anggota C



$A \cup B$ = Gabungan A dan B (or)

Hasilnya adalah semua elemen yang merupakan anggota A atau B

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Contoh Soal:

$A = \text{jumlah angka 1}$
 $A = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (3,1), (4,1), (5,1), (6,1)\}$
 $n(A) = 11$
 $P(A) = \frac{11}{36}$

$B = \text{jumlah angka 6}$
 $B = \{(1,5), (2,4), (3,3), (4,2), (5,1)\}$
 $n(B) = 5$
 $P(B) = \frac{5}{36}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $A \cap B = \{(1,5), (5,1)\}$
 $n(A \cap B) = 2$
 $P(A \cap B) = \frac{2}{36}$

$P(A \cup B) = \frac{11}{36} + \frac{5}{36} - \frac{2}{36} = \frac{14}{36}$

event ✓
 event join ✓
 sample space ✓
 rumus prob. ✓
 complement ✓
 irisan ✓
 gabungan ✓
 elemen ✓

- **Conditional probability** refers to the probability of event A , given information about the occurrence of another event, B .

CONDITIONAL PROBABILITY

The probability of A given B is equal to the probability of A and B divided by the probability of B .

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)} \quad (4.4a)$$

The probability of B given A is equal to the probability of A and B divided by the probability of A .

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)} \quad (4.4b)$$

where

$P(A \text{ and } B)$ = joint probability of A and B

$P(A)$ = marginal probability of A

$P(B)$ = marginal probability of B

2) $P(\text{Standard hanya jika streaming})$

$P(A|B) = \frac{P(A \cap B)}{P(B)}$
 $P(A \cap B) = 70/300$
 $P(B) = 108/300$
 $P(A|B) = \frac{70/300}{108/300} = \frac{70}{108}$

!!!!!!!

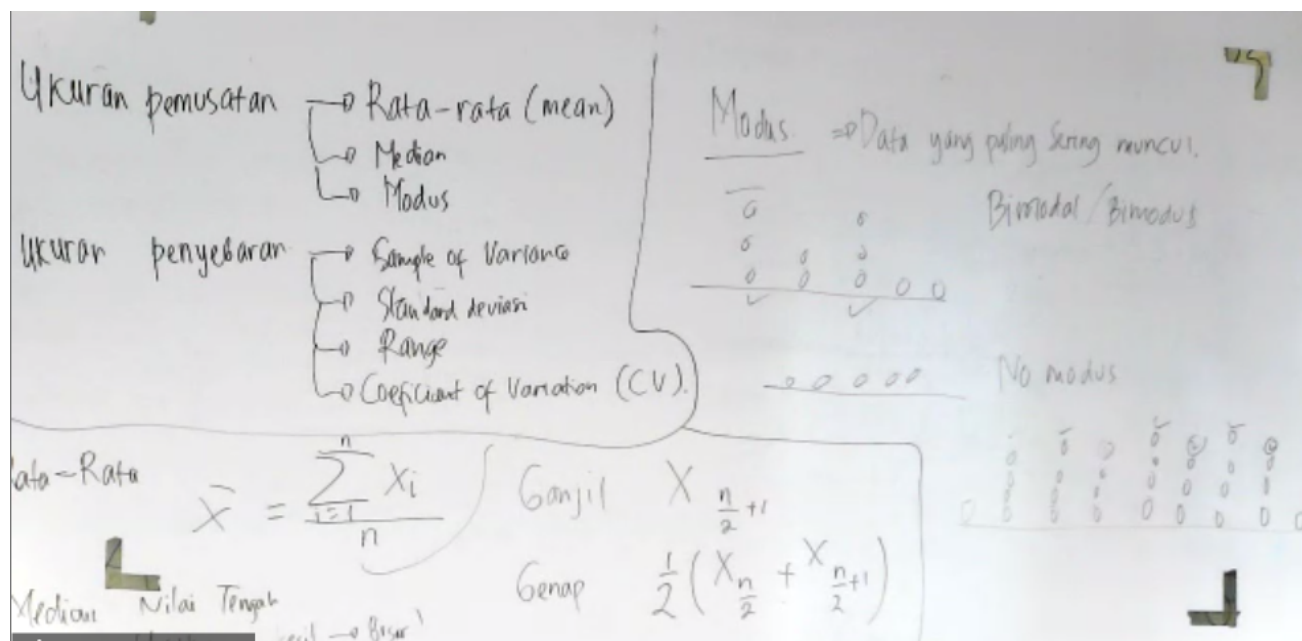
Descriptive Statistics

Metode Tabel dan Grafik

- Tabel dan Grafik data Kategori
- Tabel dan Grafik data Numerik

Metode Numerik

- Ukuran pemusatan (rata-rata, quartile, dan lain-lain)
- Ukuran penyebaran (varians, standard deviasi)



Ukuran pemusatan :

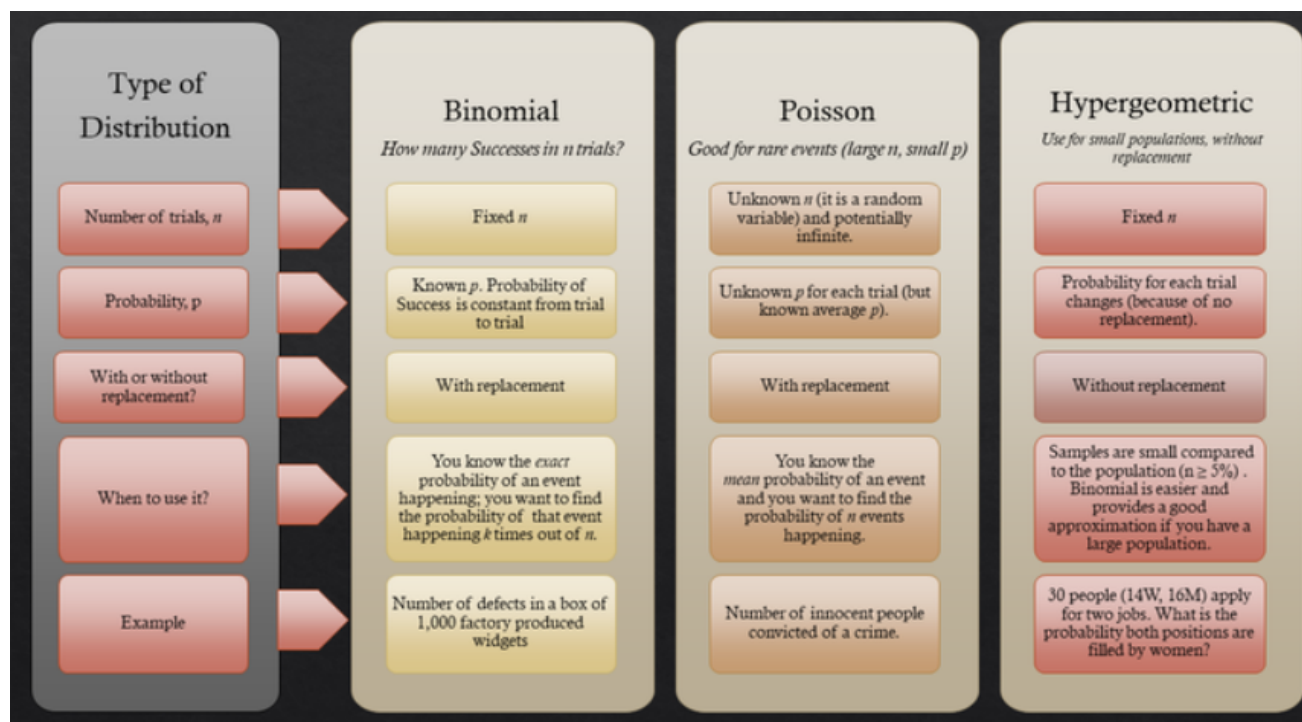
- Mean
- Median
- Modus

Ukuran penyebaran:

- Range
- Variance
- Standard Deviation
- Coefficient of Variation
- Skewness
- Varian
- Standard Deviasi
- Koefisien varian

!!!!!!!

Discrete Distribution



Binomial Distribution

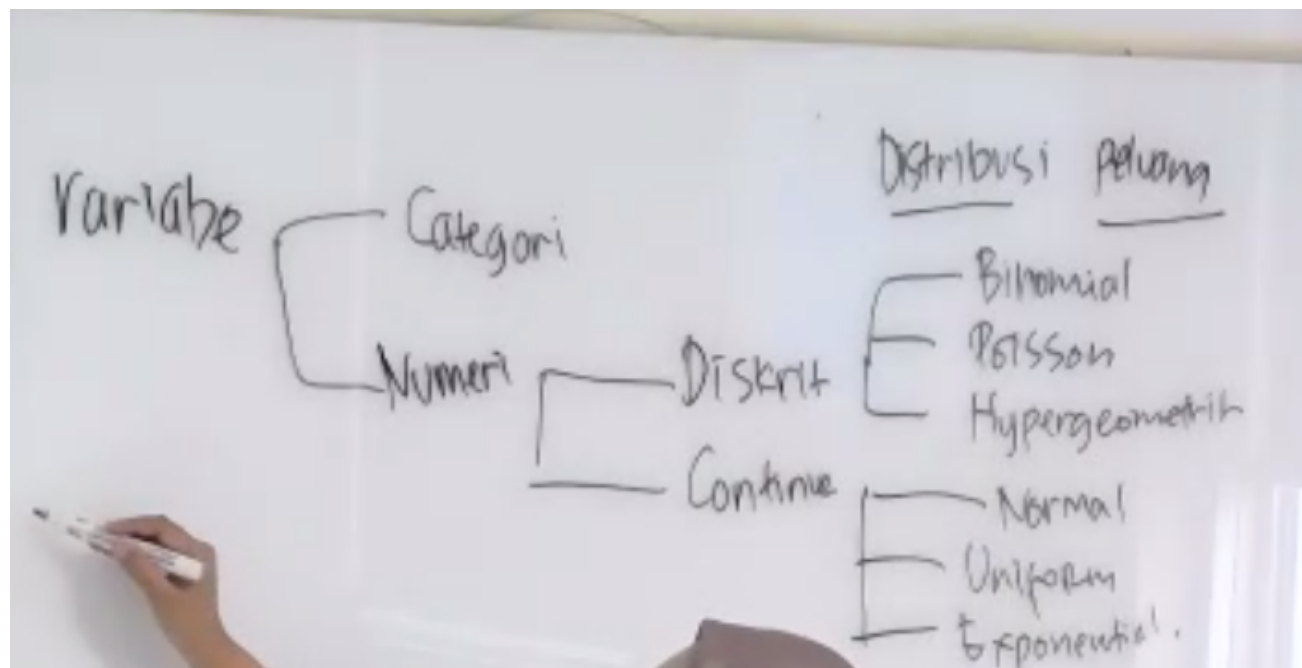
Poisson Distribution

Hypergeometric Distribution

Session 7 & 8- Normal Distribution and Continuous Distribution

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What we learn so far



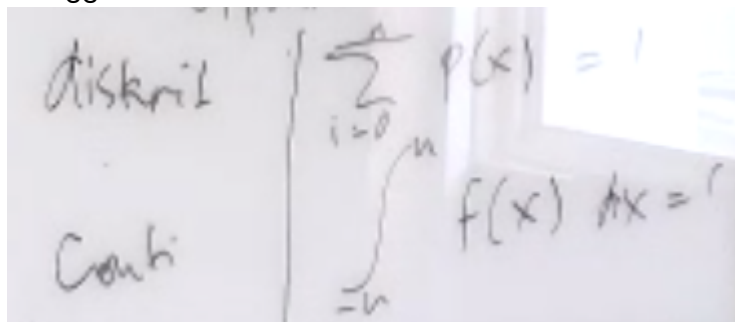
Tabel Z Distribusi Normal.pdf (kemdikbud.go.id)

Table T Distribusi Normal (ttable.org)

!!!!!!!

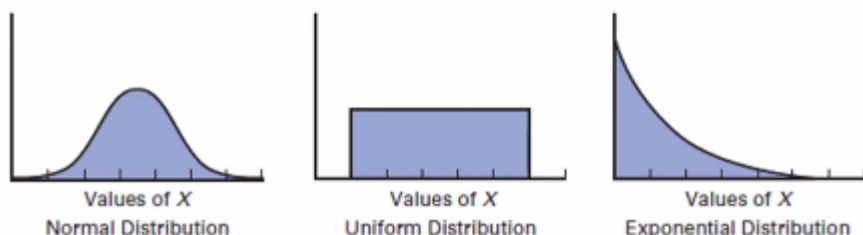
Continuous Distribution

Continuous Distribution means what being measured is not discrete (bukan jumlah), melainkan menggunakan alat ukur.



Totalnya pasti 1

A **continuous probability density function** is the mathematical expression that defines distribution of the values for a continuous random variable.



- Normal Distribution
- Uniform Distribution
- Exponential Distribution

Session 9 - Confidence Interval Estimation

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Confidence Interval

<https://www.statisticshowto.com/probability-and-statistics/confidence-interval/#:~:text=A%20confidence%20interval%20for%20the,and%207%20is%20the%20upper.>

- Zigma (Standard Deviation of Population) known and mean known

CONFIDENCE INTERVAL FOR THE MEAN (σ KNOWN)

$$\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

or

$$\bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where

$Z_{\alpha/2}$ is the value corresponding to an upper-tail probability of $\alpha/2$ from the standardized normal distribution (i.e., a cumulative area of $1 - \alpha/2$).

Step:

1. Cari $Z_{\alpha/2}$ dengan cara $(1 + \text{confidence})/2$ adalah berapa di distribusi normal?
2. Masukkan ke rumus

- Zigma (Standard Deviation of Population) unknown but mean known

CONFIDENCE INTERVAL FOR THE MEAN (σ UNKNOWN)

$$\bar{X} \pm t_{\alpha/2} \frac{S}{\sqrt{n}}$$

or

$$\bar{X} - t_{\alpha/2} \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{\alpha/2} \frac{S}{\sqrt{n}} \quad (8.2)$$

where

$t_{\alpha/2}$ is the critical value corresponding to an upper-tail probability of $\alpha/2$ (i.e., a cumulative area of $1 - \alpha/2$) from the t distribution with $n - 1$ degrees of freedom.

- For Proportion known

CONFIDENCE INTERVAL ESTIMATE FOR THE PROPORTION

$$p \pm Z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

!!!!!!!

Determining Sample Size

SAMPLE SIZE DETERMINATION FOR THE MEAN

The sample size, n , is equal to the product of the $Z_{\alpha/2}$ value squared and the standard deviation, σ , squared, divided by the square of the sampling error, e .

$$n = \frac{Z_{\alpha/2}^2 \sigma^2}{e^2} \quad (8.4)$$

sigma = standard dev populasi

Z = 1 - tail, ada di Z table distribusi normal

SAMPLE SIZE DETERMINATION FOR THE PROPORTION

The sample size n is equal to the product of $Z_{\alpha/2}$ squared, the population proportion, π , and 1 minus the population proportion, π , divided by the square of the sampling error, e .

$$n = \frac{Z_{\alpha/2}^2 \pi (1 - \pi)}{e^2} \quad (8.5)$$

pi = proporsi populasi

Z = 1 - tail, ada di Z table distribusi normal

Session 10 & 11 - Hypothesis Testing

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Hypothesis Testing

Inferential Statistic

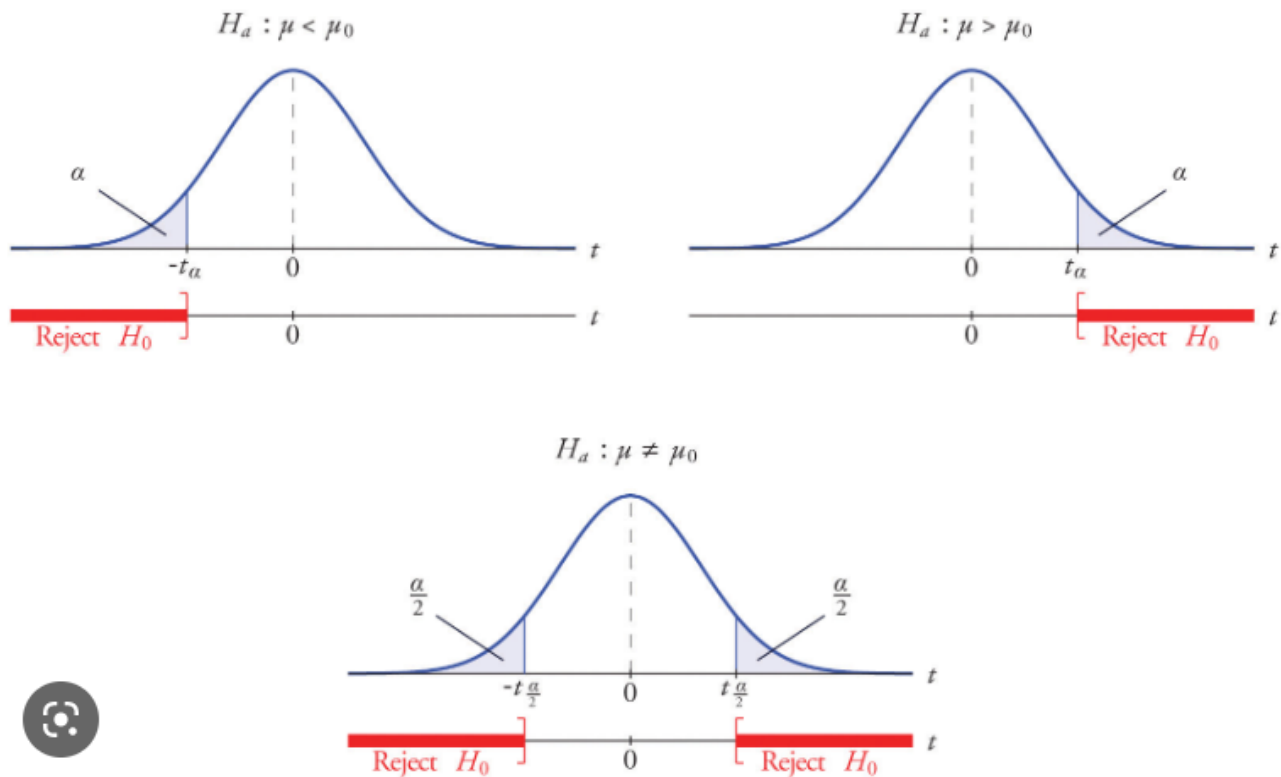
Drawing conclusions about a population by examining random samples

Skema Umum Uji Hipotesis



$$H_0 = \geq \leq$$

$$H_1 \neq < >$$



$$H_0 = H_1 \neq$$

$$H_0 \geq H_1 <$$

$$H_0 \leq H_1 >$$

Type of Hypothesis Error

Langkah:

1. Mencari Hypothesis 0 dan Hypothesis 1
2. Cari α significance level
3. Search for the critical point by $Z(\alpha)$ in the table distribution, if it's two-tailed, search for the Z from $\alpha/2$
4. Search for the Z from the case given

$$Z_{STAT} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

5. Draw the graph
6. Draw the conclusion

Untuk zigma (standard deviation of population) is unknown
Use tstat and zigma use s (standard deviation of sample)

For proportion

$$z = \frac{\hat{p} - p_o}{\sqrt{\frac{p_o(1-p_o)}{n}}}$$

Session 12 - Two Sample Tests

!!!!!!!

Two Sample Tests

POOLED-VARIANCE t TEST FOR THE DIFFERENCE BETWEEN TWO MEANS

$$t_{STAT} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad (10.1)$$

where

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{(n_1 - 1) + (n_2 - 1)}$$

and S_p^2 = pooled variance

$\bar{X}_1$ = mean of the sample taken from population 1

S_1^2 = variance of the sample taken from population 1

n_1 = size of the sample taken from population 1

$\bar{X}_2$ = mean of the sample taken from population 2

S_2^2 = variance of the sample taken from population 2

n_2 = size of the sample taken from population 2

The t_{STAT} test statistic follows a t distribution with $n_1 + n_2 - 2$ degrees of freedom.

PAIRED t TEST FOR THE MEAN DIFFERENCE

$$t_{STAT} = \frac{\bar{D} - \mu_D}{\frac{S_D}{\sqrt{n}}} \quad (10.3)$$

where

μ_D = hypothesized mean difference

$$\bar{D} = \frac{\sum_{i=1}^n D_i}{n}$$

$$S_D = \sqrt{\frac{\sum_{i=1}^n (D_i - \bar{D})^2}{n - 1}}$$

The t_{STAT} test statistic follows a t distribution with $n - 1$ degrees of freedom.

CONFIDENCE INTERVAL ESTIMATE FOR THE MEAN DIFFERENCE

$$\bar{D} \pm t_{\alpha/2} \frac{S_D}{\sqrt{n}} \quad (10.4)$$

or

$$\bar{D} - t_{\alpha/2} \frac{S_D}{\sqrt{n}} \leq \mu_D \leq \bar{D} + t_{\alpha/2} \frac{S_D}{\sqrt{n}}$$

where $t_{\alpha/2}$ is the critical value of the t distribution, with $n - 1$ degrees of freedom, for an area of $\alpha/2$ in the upper tail.

Z TEST FOR THE DIFFERENCE BETWEEN TWO PROPORTIONS

$$Z_{STAT} = \frac{(p_1 - p_2) - (\pi_1 - \pi_2)}{\sqrt{\bar{p}(1 - \bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \quad (10.5)$$

where

$$\bar{p} = \frac{X_1 + X_2}{n_1 + n_2} \quad p_1 = \frac{X_1}{n_1} \quad p_2 = \frac{X_2}{n_2}$$

and

p_1 = proportion of items of interest in sample 1

X_1 = number of items of interest in sample 1

n_1 = sample size of sample 1

π_1 = proportion of items of interest in population 1

p_2 = proportion of items of interest in sample 2

X_2 = number of items of interest in sample 2

n_2 = sample size of sample 2

π_2 = proportion of items of interest in population 2

$\bar{p}$ = pooled estimate of the population proportion of items of interest

The Z_{STAT} test statistic approximately follows a standardized normal distribution.

CONFIDENCE INTERVAL ESTIMATE FOR THE DIFFERENCE BETWEEN TWO PROPORTIONS

$$(p_1 - p_2) \pm Z_{\alpha/2} \sqrt{\frac{p_1(1 - p_1)}{n_1} + \frac{p_2(1 - p_2)}{n_2}} \quad (10.6)$$

Session 13

Session 14 -Linear Regression Model

!!!!!!!

Linear Regression Model

Dependent and Independent Variable

Mencari Linear

- Cari jumlah
- Cari rata2
- Cari SSX SSXY dll buat:
- Cari b_1
- Cari b_0
- Uji t untuk b_1
- Uji Statistik F untuk persamaan regresi



Types of Regression Models

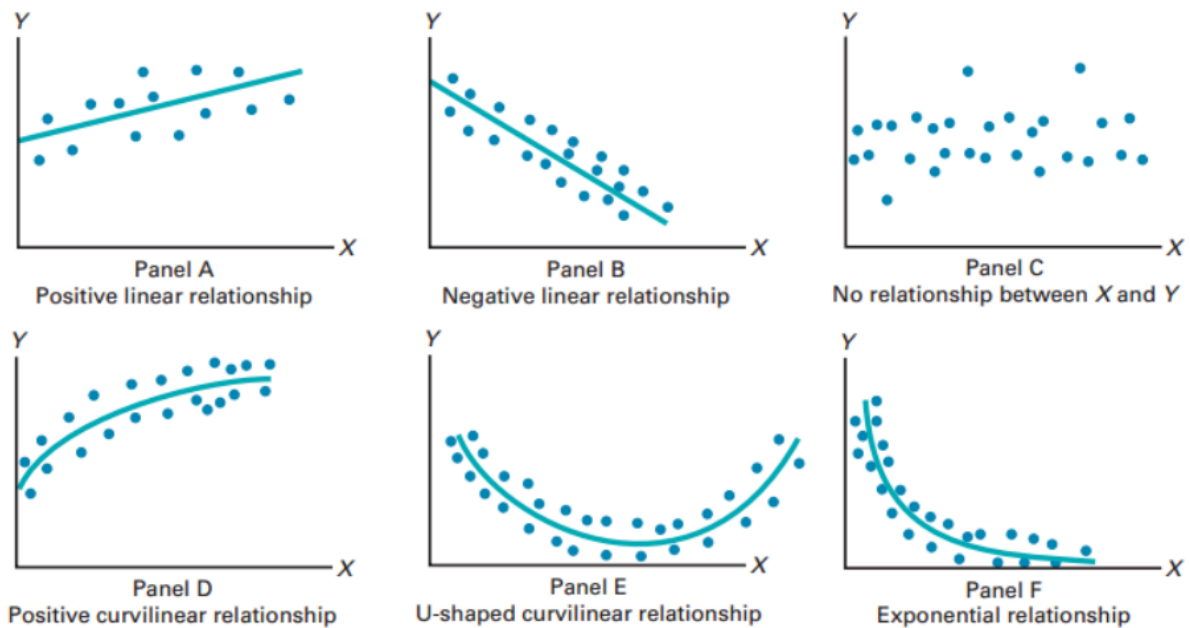


FIGURE
Six types of
relationships found
in scatter plots

Keeratan Hubungan

SIMPLE LINEAR REGRESSION MODEL

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad (13.1)$$

where

β_0 = Y intercept for the population

β_1 = slope for the population

ε_i = random error in Y for observation i

Y_i = dependent variable for observation i

X_i = independent variable for observation i

SIMPLE LINEAR REGRESSION EQUATION: THE PREDICTION LINE

The predicted value of Y equals the Y intercept plus the slope multiplied by the value of X .

$$\hat{Y}_i = b_0 + b_1 X_i \quad (13.2)$$

where

$\hat{Y}_i$ = predicted value of Y for observation i

X_i = value of X for observation i

b_0 = sample Y intercept

b_1 = sample slope

b_1 (slope)

= prediksi satu kali variabel independen x , maka akan berpengaruh berapa kali satu variabel dependen y

b_0 (sample y intercept)

= prediksi nilai awal sebelum ada variabel independen x

COMPUTATIONAL FORMULA FOR THE SLOPE, b_1

$$b_1 = \frac{SSXY}{SSX} \quad (13.3)$$

where

$$SSXY = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n X_i Y_i - \frac{\left(\sum_{i=1}^n X_i\right)\left(\sum_{i=1}^n Y_i\right)}{n}$$

$$SSX = \sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - \frac{\left(\sum_{i=1}^n X_i\right)^2}{n} \quad (\text{continued})$$

COMPUTATIONAL FORMULA FOR THE Y INTERCEPT, b_0

$$b_0 = \bar{Y} - b_1 \bar{X} \quad (13.4)$$

where

$$\bar{Y} = \frac{\sum_{i=1}^n Y_i}{n}$$

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

MEASURES OF VARIATION IN REGRESSION

The total sum of squares (SST) is equal to the regression sum of squares (SSR) plus the error sum of squares (SSE).

$$SST = SSR + SSE \quad (13.5)$$

TOTAL SUM OF SQUARES (SST)

The total sum of squares (SST) is equal to the sum of the squared differences between each observed value of Y and the mean value of Y .

$$\begin{aligned} SST &= \text{Total sum of squares} \\ &= \sum_{i=1}^n (Y_i - \bar{Y})^2 \end{aligned} \quad (13.6)$$

REGRESSION SUM OF SQUARES (SSR)

The regression sum of squares (SSR) is equal to the sum of the squared differences between each predicted value of Y and the mean value of Y .

$$\begin{aligned} SSR &= \text{Explained variation or regression sum of squares} \\ &= \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 \end{aligned} \quad (13.7)$$

ERROR SUM OF SQUARES (SSE)

The error sum of squares (SSE) is equal to the sum of the squared differences between each observed value of Y and the predicted value of Y .

$$\begin{aligned} SSE &= \text{Unexplained variation or error sum of squares} \\ &= \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 \end{aligned} \quad (13.8)$$

SOLUTION You can compute SST , SSR , and SSE , which are defined in Equations (13.6), (13.7), and (13.8) on pages 437 and 438, by using Equations (13.10), (13.11), and (13.12).

COMPUTATIONAL FORMULA FOR SST

$$SST = \sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n Y_i^2 - \frac{\left(\sum_{i=1}^n Y_i\right)^2}{n} \quad (13.10)$$

COMPUTATIONAL FORMULA FOR SSR

$$\begin{aligned} SSR &= \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 \\ &= b_0 \sum_{i=1}^n Y_i + b_1 \sum_{i=1}^n X_i Y_i - \frac{\left(\sum_{i=1}^n Y_i\right)^2}{n} \end{aligned} \quad (13.11)$$

COMPUTATIONAL FORMULA FOR SSE

$$SSE = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n Y_i^2 - b_0 \sum_{i=1}^n Y_i - b_1 \sum_{i=1}^n X_i Y_i \quad (13.12)$$

COEFFICIENT OF DETERMINATION

The coefficient of determination is equal to the regression sum of squares (i.e., explained variation) divided by the total sum of squares (i.e., total variation).

$$r^2 = \frac{\text{Regression sum of squares}}{\text{Total sum of squares}} = \frac{SSR}{SST} \quad (13.9)$$

$r > 0$ – korelasi positif --> makin besar nilai X maka Y juga semakin besar
X makin rendah Y juga makin rendah
$r < 0$ – korelasi negatif --> makin besar X maka Y makin menurun
makin rendah X maka Y semakin meningkat

$r^2 = \frac{SSR}{SST} = 0,832731$

Keragaman/variasi data (var Y) dapat diterangkan oleh persamaan regresi sebesar 83,27 persen

Koefisien Korelasi r

Menduga tingkat keeratan dan arah hubungan antara 2 variabel x dan y

Koefisien Determinasi r^2

Keragaman/variasi data (var y) diterangkan oleh persamaan regresi

The standard error of the estimate is the standard deviation around the prediction line, whereas the standard deviation in Chapter 3 is the standard deviation around the sample mean. Equation (13.13) defines the standard error of the estimate, represented by the symbol S_{YX} .

STANDARD ERROR OF THE ESTIMATE

$$S_{YX} = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{n-2}} \quad (13.13)$$

where

Y_i = observed value of Y for a given X_i

$\hat{Y}_i$ = predicted value of Y for a given X_i

SSE = error sum of squares

From Equation (13.8) and Figure 13.4 or Figure 13.7, $SSE = 11.9832$. Thus,

$$S_{YX} = \sqrt{\frac{11.9832}{14-2}} = 0.9993$$

The four assumptions of regression (known by the acronym LINE) are:

- Linearity
- Independence of errors
- Normality of error
- Equal variance

DURBIN-WATSON STATISTIC

$$D = \frac{\sum_{i=2}^n (e_i - e_{i-1})^2}{\sum_{i=1}^n e_i^2} \quad (13.15)$$

where

e_i = residual at the time period i



t Test for the Slope

To determine the existence of a significant linear relationship between the X and Y variables, you test whether b_1 (the population slope) is equal to 0. The null and alternative hypotheses are as follows:

$$\begin{aligned} H_0: \beta_1 &= 0 \text{ [There is no linear relationship (the slope is zero).]} \\ H_1: \beta_1 &\neq 0 \text{ [There is a linear relationship (the slope is not zero).]} \end{aligned}$$

TESTING A HYPOTHESIS FOR A POPULATION SLOPE, β_1 , USING THE t TEST

The t_{STAT} test statistic equals the difference between the sample slope and hypothesized value of the population slope divided by S_{b_1} , the standard error of the slope.

$$t_{STAT} = \frac{b_1 - \beta_1}{S_{b_1}} \quad (13.16)$$

where

$$S_{b_1} = \frac{S_{YX}}{\sqrt{SSX}}$$

$$SSX = \sum_{i=1}^n (X_i - \bar{X})^2$$

The t_{STAT} test statistic follows a t distribution with $n - 2$ degrees of freedom.

TESTING A HYPOTHESIS FOR A POPULATION SLOPE, β_1 , USING THE F TEST

The F_{STAT} test statistic is equal to the regression mean square (MSR) divided by the mean square error (MSE).

$$F_{STAT} = \frac{MSR}{MSE} \quad (13.17)$$

where

$$MSR = \frac{SSR}{1} = SSR$$

$$MSE = \frac{SSE}{n - 2}$$

The F_{STAT} test statistic follows an F distribution with 1 and $n - 2$ degrees of freedom.

CONFIDENCE INTERVAL ESTIMATE OF THE SLOPE, β_1

The confidence interval estimate for the population slope can be constructed by taking the sample slope, b_1 , and adding and subtracting the critical t value multiplied by the standard error of the slope.

$$\begin{aligned} b_1 \pm t_{\alpha/2} S_{b_1} \\ b_1 - t_{\alpha/2} S_{b_1} \leq \beta_1 \leq b_1 + t_{\alpha/2} S_{b_1} \end{aligned} \quad (13.18)$$

where

$t_{\alpha/2}$ = critical value corresponding to an upper-tail probability of $\alpha/2$ from the t distribution with $n - 2$ degrees of freedom (i.e., a cumulative area of $1 - \alpha/2$)

TESTING FOR THE EXISTENCE OF CORRELATION

$$t_{STAT} = \frac{r - \rho}{\sqrt{\frac{1 - r^2}{n - 2}}} \quad (13.19a)$$

where

$$r = +\sqrt{r^2} \quad \text{if } b_1 > 0$$

$$r = -\sqrt{r^2} \quad \text{if } b_1 < 0$$

The t_{STAT} test statistic follows a t distribution with $n - 2$ degrees of freedom. r is calculated as in Equation (3.17) on page 126:

$$r = \frac{\text{cov}(X, Y)}{S_X S_Y} \quad (13.19b)$$

where

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

$$S_X = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}}$$

$$S_Y = \sqrt{\frac{\sum_{i=1}^n (Y_i - \bar{Y})^2}{n - 1}}$$

!!!!!!!

BasicStat ANKI

TARGET DECK: Basic Statistics

What is the difference of Mean and Proportion? #flashcard

Mean is used to search for average in numerical data, so the answer would be roughly the middle of the numerical data, it's probably more than 1, depending on what's tracked.

Proportion is used to search for the amount of successes in binomial data, it always is between 0 and 1. Proportion is like the probability of success.

Mean = sum/N

Proportion = successes/N

8.18 The file **FastFood** contains the amount that a sample of 15 customers spent for lunch (\$) at a fast-food restaurant:

7.42 6.29 5.83 6.50 8.34 9.51 7.10 6.80 5.90
4.89 6.50 5.52 7.90 8.30 9.60

- Construct a 95% confidence interval estimate for the population mean amount spent for lunch (\$) at a fast-food restaurant, assuming a normal distribution.
- Interpret the interval constructed in (a).

#flashcard

-tbd-

8.23 A manufacturing company produces electric insulators. You define the variable of interest as the strength of the insulators. If the insulators break when in use, a short circuit is likely. To test the strength of the insulators, you carry out destructive testing to determine how much force is required to break the insulators. You measure force by observing how many pounds are applied to the insulator before it breaks. You collect the force data for 30 insulators selected for the experiment and organize and store these data in **Force** :

1,870 1,728 1,656 1,610 1,634 1,784 1,522 1,696
1,592 1,662 1,866 1,764 1,734 1,662 1,734 1,774
1,550 1,756 1,762 1,866 1,820 1,744 1,788 1,688
1,810 1,752 1,680 1,810 1,652 1,736

- Construct a 95% confidence interval estimate for the population mean force.
- What assumption must you make about the population distribution in order to construct the confidence interval estimate in (a)?
- Do you think that the assumption needed in order to construct the confidence interval estimate in (a) is valid? Explain.

#flashcard

-tbd-

What is a type I error? #flashcard

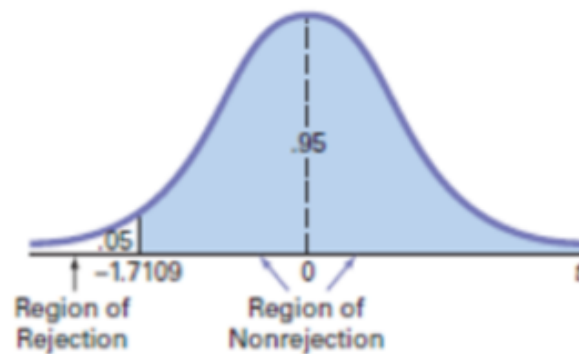
Rejecting the correct hypothesis

What is a type II error? #flashcard

Accepting the wrong hypothesis

One example of a one-tail hypothesis test would test whether the population mean is *less than* a specified value. One such situation involves the business problem concerning the service time at the drive-through window of a fast-food restaurant. According to a *QSR* magazine special report (“The Drive-Thru Study,” October 2013, available at bit.ly/1IU6Enz), the speed with which customers are served is of critical importance to the success of the service. In that study, an audit of McDonald’s drive-throughs had a mean service time of 189.49 seconds, which was slower than the drive-throughs of several other fast-food chains. Suppose that McDonald’s began a quality improvement effort to reduce the service time by deploying an improved drive-through service process in a sample of 25 stores. Because McDonald’s would want to institute the new process in all of its stores only if the test sample saw a *decreased* drive-through time, the entire rejection region is located in the lower tail of the distribution.

Do complete 1-6 steps! #flashcard



Step 5 From the sample of 25 stores you selected, you find that the sample mean service time at the drive-through equals 175.8 seconds and the sample standard deviation equals 21.3 seconds. Using $n = 25$, $\bar{X} = 175.8$, $S = 21.3$, and Equation (9.2) on page 308,

$$t_{STAT} = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} = \frac{175.8 - 189.49}{\frac{21.3}{\sqrt{25}}} = -3.2136$$

Step 6 Because $t_{STAT} = -3.2136 < -1.7109$, you reject the null hypothesis (see Figure 9.10). You conclude that the mean service time at the drive-through is less than 189.49 seconds. There is sufficient evidence to change the drive-through process for the entire population of stores.

A company that manufactures chocolate bars is particularly concerned that the mean weight of a chocolate bar is not greater than 6.03 ounces. A sample of 50 chocolate bars is selected; the sample mean is 6.034 ounces, and the sample standard deviation is 0.02 ounce. Using the $\alpha = 0.01$ level of significance, is there evidence that the population mean weight of the chocolate bars is greater than 6.03 ounces?

Answer the question step by step! #flashcard

SOLUTION Using the critical value approach, listed in Exhibit 9.1 on page 301,

Step 1 First, you define the null and alternative hypotheses:

$$H_0 : \mu \leq 6.03$$

$$H_1 : \mu > 6.03$$

Step 2 You collect the data from a sample of $n = 50$. You decide to use $\alpha = 0.01$.

Step 3 Because σ is unknown, you use the t distribution and the t_{STAT} test statistic.

Step 4 The rejection region is entirely contained in the upper tail of the sampling distribution of the mean because you want to reject H_0 only when the sample mean is significantly greater than 6.03 ounces. Because the entire rejection region is in the upper tail of the t distribution and contains an area of 0.01, the critical value of the t distribution with $50 - 1 = 49$ degrees of freedom is 2.4049 (see Table E.3).

The decision rule is

reject H_0 if $t_{STAT} > 2.4049$;

otherwise, do not reject H_0 .

Step 5 From your sample of 50 chocolate bars, you find that the sample mean weight is 6.034 ounces, and the sample standard deviation is 0.02 ounces. Using $n = 50$, $\bar{X} = 6.034$, $S = 0.02$, and Equation (9.2) on page 308,

$$t_{STAT} = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} = \frac{6.034 - 6.03}{\frac{0.02}{\sqrt{50}}} = 1.414$$

Step 6 Because $t_{STAT} = 1.414 < 2.4049$ or the p -value (from Excel) is $0.0818 > 0.01$, you do not reject the null hypothesis. There is insufficient evidence to conclude that the population mean weight is greater than 6.03 ounces.

You are the manager of a fast-food restaurant. The business problem is to determine whether the population mean waiting time to place an order has changed in the past month from its previous value of 4.5 minutes. From past experience, you can assume that the population standard deviation is 1.2 minutes and the population waiting time is normally distributed. You select a sample of 25 orders during a one-hour period. The sample mean is 5.1 minutes. Use the five-step p -value approach of Exhibit 9.2 to determine whether there is evidence that the population mean waiting time to place an order has changed in the past month from its previous population mean value of 4.5 minutes.

Use p -value AND critical value method to prove they both got the same conclusion! #flashcard

Step 4 You collect the sample data and compute $\bar{X} = 5.1$. Using Equation (9.1) on page 300, you compute the test statistic as follows:

$$Z_{STAT} = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{5.1 - 4.5}{\frac{1.2}{\sqrt{25}}} = +2.50$$

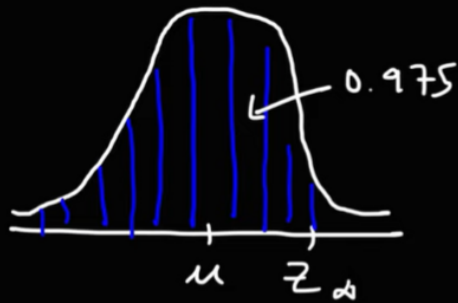
To find the probability of getting a Z_{STAT} test statistic that is equal to or more extreme than 2.50 standard error units from the center of a standardized normal distribution, you compute the probability of a Z_{STAT} value greater than +2.50 along with the probability of a Z_{STAT} value less than -2.50. From Table E.2, the probability of a Z_{STAT} value below -2.50 is 0.0062. The probability of a value below +2.50 is 0.9938. Therefore, the probability of a value above +2.50 is $1 - 0.9938 = 0.0062$. Thus, the p -value for this two-tail test is $0.0062 + 0.0062 = 0.0124$.

Step 5 Because the p -value = 0.0124 < α = 0.05, you reject the null hypothesis. You conclude that there is evidence that the population mean waiting time to place an order has changed from its previous population mean value of 4.5 minutes. The mean waiting time for customers is longer now than it was last month.

1. The average test scores in a physics class is normally distributed with a standard deviation of 5.4. 50 scores with a sample mean of 79 were selected at random. (a) Find a 95% confidence interval for the population mean test score. (b) What is the value of the margin of error? i

Do question a and b #flashcard

1. The average test scores in a physics class is normally distributed with a standard deviation of 5.4. 50 scores with a sample mean of 79 were selected at random. (a) Find a 95% confidence interval for the population mean test score. (b) What is the value of the margin of error?



$$\sigma = 5.4 \quad n = 50 \quad \bar{X} = 79$$

$$\bar{X} - z_{\alpha} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha} \frac{\sigma}{\sqrt{n}}$$

$$CI \rightarrow (\bar{X} - EBM, \bar{X} + EBM)$$

$$CI \rightarrow \bar{X} \pm z_{\alpha} \frac{\sigma}{\sqrt{n}}$$

$$z_{\alpha} = 1.96$$

$$95\% CI \rightarrow 79 \pm 1.96(0.76368)$$

$$79 \pm 1.4968$$

$$(77.5032, 80.4968)$$

The average test scores in a chemistry class is normally distributed with a standard deviation of 6.5. 100 scores with a sample mean of 82 were selected at random. Find a 90% confidence interval for the population mean exam score and what is the value of the margin of error?

#flashcard

2. The average test scores in a chemistry class is normally distributed with a standard deviation of 6.5. 100 scores with a sample mean of 82 were selected at random. (a) Find a 90% confidence interval for the population mean exam score. (b) What is the value of the margin of error?

$$\sigma = 6.5 \quad n = 100 \quad \bar{X} = 82$$

$$\bar{X} \pm z_{\alpha} \frac{\sigma}{\sqrt{n}}$$

$$A_c = \frac{CL + 1}{2} = \frac{0.90 + 1}{2} = 0.95 \rightarrow z_{\alpha} = 1.645$$

$$82 \pm 1.645 \left(\frac{6.5}{\sqrt{100}} \right)$$

$$EBM = z_{\alpha} \frac{\sigma}{\sqrt{n}}$$

$$82 \pm 1.06925$$

$$1.645 \left(\frac{6.5}{\sqrt{100}} \right)$$

$$(80.93075, 83.06925)$$

$$= 1.06925$$

Example: Consider the distribution of serum cholesterol levels for all males in the US who are hypertensive and who smoke. This distribution has an unknown mean μ and a standard deviation 46 mg/100ml. Suppose we draw a random sample of 12 individual from this population and find that the mean cholesterol level is $\bar{x} = 217\text{mg}/100\text{ml}$.

For 95% confidence, what is the confidence interval? #flashcard
(191, 243)

Example: Consider the distribution of serum cholesterol levels for all males in the US who are hypertensive and who smoke. This distribution has an unknown mean μ and a standard deviation 46 mg/100ml. Suppose we draw a random sample of 12 individual from this population and find that the mean cholesterol level is $\bar{x} = 217\text{mg}/100\text{ml}$.

For 95% confidence, how large a sample would be needed to reduce the margin of error to be 20? #flashcard

Recall that the 95% confidence interval is

$$\left(217 - 1.96 \frac{46}{\sqrt{n}}, 217 + 1.96 \frac{46}{\sqrt{n}} \right).$$

The length of this confidence interval is $2 \times 1.96 \times 46 / \sqrt{n}$. So to find the required sample size n , we can solve the equation

$$2 \times 1.96 \frac{46}{\sqrt{n}} = 20.$$

We find

$$n = \left(\frac{1.96 \times 46}{10} \right)^2 = 81.3 \approx 82.$$

Example: A sample of 16 ten-year-old girls gave a mean weight of 71.5 and a standard deviation of 12 pounds. Assuming normality, find the 90, 95, and 99 percent confidence intervals for the population mean weight μ .

Semangat kak! #flashcard

Let $t_{\alpha/2}(n - 1)$ the value that cuts off the upper area of $\alpha/2$ in a t-distribution with $n - 1$ degrees of freedom. The general form of the confidence interval based on the t-distribution is

$$\left(\bar{x} - t_{\alpha/2}(n - 1) \frac{s}{\sqrt{n}}, \quad \bar{x} + t_{\alpha/2}(n - 1) \frac{s}{\sqrt{n}} \right).$$

The 90, 95, and 99 percent confidence intervals are

$$\left(71.5 - 1.75 \frac{12}{4}, \quad 71.5 + 1.75 \frac{12}{4} \right),$$

$$\left(71.5 - 2.13 \frac{12}{4}, \quad 71.5 + 2.13 \frac{12}{4} \right),$$

$$\left(71.5 - 2.95 \frac{12}{4}, \quad 71.5 + 2.95 \frac{12}{4} \right),$$

or

$$(66.25, \quad 76.75),$$

$$(65.11, \quad 77.89),$$

$$(62.65, \quad 80.35),$$

Returning to the cereal-filling process described in the Using Statistics scenario on page 240, if you randomly select a sample of 25 boxes without replacement from the thousands of boxes filled during a shift, the sample contains a very small portion of the population. Given that the standard deviation of the cereal-filling process is 15 grams, compute the standard error of the mean.

smgt pasti bisa! #flashcard

SOLUTION Using Equation (7.3) with $n = 25$ and $\sigma = 15$ the standard error of the mean is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{25}} = \frac{15}{5} = 3$$

The variation in the sample means for samples of $n = 25$ is much less than the variation in the individual boxes of cereal (i.e., $\sigma_{\bar{x}} = 3$, while $\sigma = 15$).

Berdasarkan 100 laporan kejadian hujan (dengan lama kejadian hujan sama) di daerah "SH" yang diamati secara acak, diperoleh bahwa rata-rata tingkat curah hujan adalah 71,8 mm dengan simpangan baku 8,9 mm. Berdasarkan literatur diduga bahwa rata-rata tingkat curah hujan di daerah tersebut lebih dari 70 mm.

- Nyatakan dugaan tersebut dalam pernyataan hipotesis statistik
- Untuk tingkat signifikansi 5%, benarkah pernyataan literatur tersebut?

Apakah H0 terbukti? #flashcard

Solusi

Diketahui

Ditanya:

- Hipotesis statistik
- Kesimpulan uji hipotesis

Jawab:

Parameter yang akan diuji : μ

a. Rumusan hipotesis:

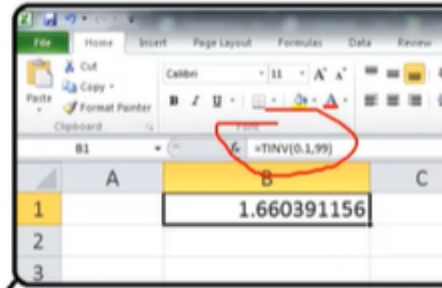
$$H_0: \mu = 70$$

$$H_1: \mu > 70$$

- b. $\alpha = 5\% = 0.05$, maka titik kritis $t_{0.05, (99)} = 1.645$

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{71.8 - 70}{8.9/\sqrt{100}} = 2.02$$

- Karena $t > t_{0.05, (99)}$, maka t berada pada daerah penolakan sehingga keputusannya H_0 ditolak.
- Jadi sampel yang ada mendukung pernyataan literatur tersebut, yaitu bahwa rata-rata tingkat curah hujan di daerah "SH" lebih dari 70 mm.



Ini dari manakah?

$$\begin{aligned}\frac{99 - 80}{100 - 80} &= \frac{x - 1.664}{1.660 - 1.664} \\ 0.95 &= \frac{x - 1.664}{-0.004} \\ -0.0038 &= x - 1.664 \\ 1.6602 &= x\end{aligned}$$



Example

Berdasarkan 100 laporan kejadian hujan (dengan lama kejadian hujan sama) di daerah "SH" yang diamati secara acak, diperoleh bahwa rata-rata tingkat curah hujan adalah 71,8 mm dengan simpangan baku 8,9 mm. Berdasarkan literatur diduga bahwa rata-rata tingkat curah hujan di daerah tersebut lebih dari 70 mm.

- Nyatakan dugaan tersebut dalam pernyataan hipotesis statistik
- Untuk tingkat signifikansi 5%, benarkah pernyataan literatur tersebut?

People
Innovation
Excellence

Apakah H_0 terbukti? #flashcard

Solusi

Diketahui

Ditanya:

- Hipotesis statistik
- Kesimpulan uji hipotesis

Jawab:

Parameter yang akan diuji : μ

a. Rumusan hipotesis:

$$H_0: \mu = 70$$

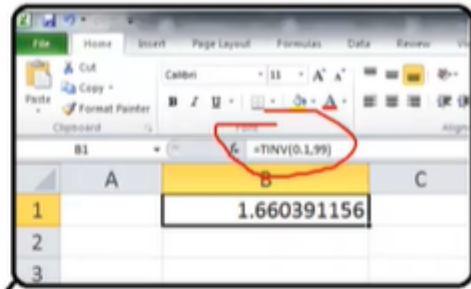
$$H_1: \mu > 70$$

- b. $\alpha = 5\% = 0.05$, maka titik kritis $t_{0.05, (99)} = 1.645$

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} = \frac{71.8 - 70}{8.9 / \sqrt{100}} = 2.02$$

- Karena $t > t_{0.05, (99)}$, maka t berada pada daerah penolakan sehingga keputusannya H_0 ditolak.
- Jadi sampel yang ada mendukung pernyataan literatur tersebut, yaitu bahwa rata-rata tingkat curah hujan di daerah "SH" lebih dari 70 mm.

Ini dari manakah?



$$\begin{aligned} \frac{99 - 80}{100 - 80} &= \frac{x - 1.664}{1.660 - 1.664} \\ 0.95 &= \frac{x - 1.664}{-0.004} \\ -0.0038 &= x - 1.664 \\ 1.6602 &= x \end{aligned}$$

What is the formula for pooled variance t test for the difference between two means? #flashcard

POOLED-VARIANCE t TEST FOR THE DIFFERENCE BETWEEN TWO MEANS

$$t_{STAT} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad (10.1)$$

where

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{(n_1 - 1) + (n_2 - 1)}$$

You and some friends have decided to test the validity of an advertisement by a local pizza restaurant, which says it delivers to the dormitories faster than a local branch of a national chain. Both the local pizza restaurant and national chain are located across the street from your college campus. You define the variable of interest as the delivery time, in minutes, from the time the pizza is ordered to when it is delivered. You collect the data by ordering 10 pizzas from the local pizza restaurant and 10 pizzas from the national chain at different times. You organize and store the data in `PizzaTime`. Table 10.2 shows the delivery times.

Local		Chain	
16.8	18.1	22.0	19.5
11.7	14.1	15.2	17.0
15.6	21.8	18.7	19.5
16.7	13.9	15.6	16.5
17.5	20.8	20.8	24.0

At the 0.05 level of significance, is there evidence that the mean delivery time for the local pizza restaurant is less than the mean delivery time for the national pizza chain?

SOLUTION Because you want to know whether the mean is *lower* for the local pizza restaurant than for the national pizza chain, you have a one-tail test with the following null and alternative hypotheses:

$H_0: \mu_1 \geq \mu_2$ (The mean delivery time for the local pizza restaurant is equal to or greater than the mean delivery time for the national pizza chain.)

$H_1: \mu_1 < \mu_2$ (The mean delivery time for the local pizza restaurant is less than the mean delivery time for the national pizza chain.)

Figure 10.5 displays the results for the pooled-variance t test for these data.

To illustrate the computations, using Equation (10.1) on page 332,

$$t_{STAT} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

where

$$\begin{aligned} S_p^2 &= \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{(n_1 - 1) + (n_2 - 1)} \\ &= \frac{9(3.0955)^2 + 9(2.8662)^2}{9 + 9} = 8.8986 \end{aligned}$$

Therefore,

$$t_{STAT} = \frac{(16.7 - 18.88) - 0.0}{\sqrt{8.8986 \left(\frac{1}{10} + \frac{1}{10} \right)}} = \frac{-2.18}{\sqrt{1.7797}} = -1.6341$$

An online survey asked 1,004 adults “If purchasing a used car made certain upgrades or features more affordable, what would be your preferred luxury upgrade?” The results indicated that 9% of the males and 14% of the females answered window tinting. (Source: Data extracted from Ipsos, “Safety Technology Tops the List of Most Desired Features Should They Be More Affordable when Purchasing a Used Car—Particularly Collision Avoidance,” bit.ly/1RCcc1L.) The sample sizes of males and females were not provided. Suppose that both sample sizes were 502 and that 46 of 502 males and 71 of 502 females reported window tinting as their preferred luxury upgrade of choice.

- a. Is there evidence of a difference between males and females in the proportion who said they prefer window tinting as a luxury upgrade at the 0.01 level of significance?
- b. Find the p -value in (a) and interpret its meaning.
- c. Construct and interpret a 99% confidence interval estimate for the difference between the proportion of males and females who said they prefer window tinting as a luxury upgrade.
- d. What are your answers to (a) through (c) if 60 males said they prefer window tinting as a luxury upgrade?

Answer semangatt! #flashcard

-tbd-